

Profile of First Time Seizure in Infants with 1 to 12 Months of Age Presenting to a Tertiary Care Pediatric Hospital

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Abstract

The incidence of seizure in first year of life is very high. Considerable research on neonatal seizures has been done; however information regarding the profile of patients with first time seizure in the age group of 1 to 12 months is scarce. Our study was a prospective observational study undertaken at a tertiary pediatric care hospital in north India with objectives of finding the epidemiological, clinical, and aetiological profile of patients in the age group of 1 to 12 months presenting with first time seizure and to ascertain their electroencephalography (EEG) and imaging findings. Seizure recurrence and developmental delay was also studied during the first six months post seizure. Fifty patients were included in the study. There were 31 male patients (62%) and 19 female patients (38%). Mean age at seizure occurrence was 5.3 ± 2.1 months. Forty one patients (82%) presented with generalized tonic seizures. Only two patients (4%) had focal seizures. Seizures in 7 patients (14%) could not be classified. Electroencephalography and brain imaging were done in 42 patients as indicated. Among these 42 patients, EEG abnormalities were noted in 15 patients (35.7%), while imaging abnormalities were noted in 10 patients (23%). Fifteen patients (35.7%) were found to have developmental delay on follow-up. 16 (38%) patients developed recurrence of seizures. Statistical analysis revealed significant association between abnormal imaging and developmental delay at six months and seizure recurrence within six months of first seizure, however the association with abnormal EEG was not statistically significant.

Keywords

- infant seizures
- developmental delay
- seizure recurrence
- EEG
- imaging

Introduction

The incidence of seizure in first year of life is very high.¹ There is a sharp difference between neonatal and postneonatal seizures in infants with respect to etiology, semiology, electroencephalography (EEG), and imaging findings and final prognosis.² Considerable research on neonatal seizures has been done; however, information regarding the profile of patients with first time seizure in the age group of 1 to 12 months is scarce. Our study was undertaken with objectives of finding the epidemiological, clinical, and etiological profile of patients in the age group of 1 to 12 months presenting

with first time seizure to our hospital and to ascertain their EEG and imaging findings. Seizure recurrence and developmental delay was also studied during the first 6 months postseizure.

Materials and Methods

This was an observational prospective study conducted at the GB Pant Children Hospital, Srinagar, a tertiary care hospital for children from March 2016 to December 2016. A written informed consent was obtained from parents or guardians. Study population included infants between age

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group of 1 to 12 months presenting to the hospital with first time seizure. Infants with any significant perinatal history (history of neonatal jaundice needing phototherapy or exchange transfusion, birth asphyxia, hypoglycemia, or with prior history of seizures) were excluded from the study. History was taken about the seizure from guardians to differentiate between actual seizure and seizure mimics and to ascertain the semiology of seizure event. Attendants were encouraged to video image the seizure if recurrent so as to be certain about the seizure occurrence. A detailed history was also sought regarding other variables relevant to the study including a thorough developmental history. A detailed examination was done under controlled conditions after patient stabilization including anthropometry, general physical and systemic examination, and developmental examination. A child psychologist was involved in developmental assessment. Development Assessment Scale for Indian Infants (DASII) was used in this study. This scale is based on Bayley Scale of Infant Development and is used to assess development of children from birth through 2 years of age. It measures motor development and mental development as motor developmental quotient (DQ) and mental DQ. The normative values of DASII are well established and this scale is being used frequently in Indian studies.^{3–5} A DQ score of < 70 (< 2 standard deviation) in motor and/or mental DQ is taken as developmental delay. Stigmata of intrauterine infections and any neurocutaneous markers were sought. Assessment for hearing and visual impairment was also done. Patients received treatment as per standard management protocols. Relevant investigations (blood sugar, serum total calcium, serum electrolytes, and septic workup) were done during hospital stay as required. An EEG was done after 2 weeks of seizure occurrence wherever indicated. Given the large number of patients who undergo EEG on daily basis at our EEG facility, we were not in a position to do EEG as soon as possible which is most desirable; however, we still worked it out at 2 weeks for all patients so that at least recommended period of within 4 weeks is not missed. EEG was read by a neurologist and for the purposes of this study, the findings were recorded as either normal or abnormal if any of the epileptiform discharges were present. Specific epileptiform abnormalities were not studied. Brain imaging, tandem mass spectrometry, and urine gas chromatography–mass spectrometry were also advised as required according to protocols. Patients were followed up for 6 months after initial presentation with seizure and contact with primary care taker was maintained through telephonic calls and planned outpatient visits. For statistical analysis, Fisher's exact test was used for discrete variables and Student's *t*-test for continuous variables. Relative risk (RR) was also calculated wherever required.

Results

Out of 177 consecutive patients (102 males and 75 females) presenting as first time seizure after neonatal age during the study period, 76 (42.9%) presented within first year of life. Twenty infants had significant history during neonatal per-

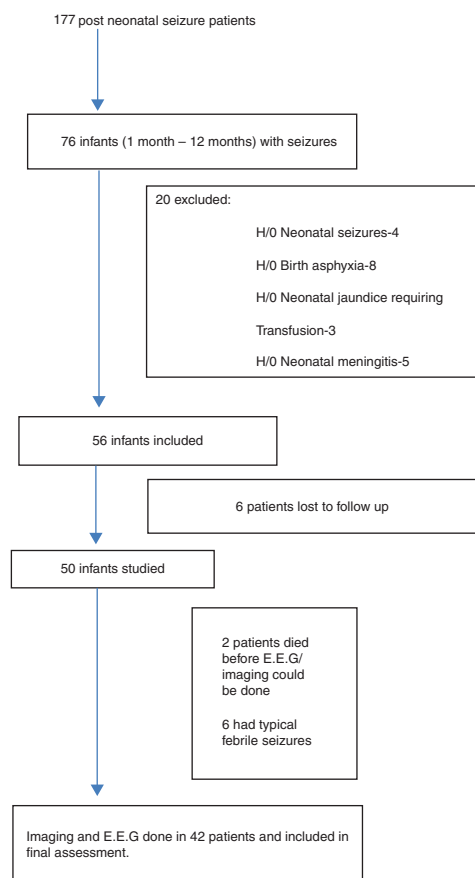


Fig. 1 Flowchart of participants.

iod and were excluded. Six patients were lost to follow-up. Fifty patients were included in the study (**► Fig. 1**): 31 male patients (62%) and 19 female patients (38%). Mean age at seizure occurrence was 5.3 ± 2.1 months, with maximum age at presentation of 12 months and minimum of 1.5 months. All infants in the study sample were born at term. Although preterm delivery was not kept as exclusion criteria, no such infant was encountered during our study, likely in view of small study sample. Seizures were observed in hospital in 28 patients by the emergency doctor (a trained pediatrician), while in 10 patients, the event was captured by guardians on mobile phone camera. In the remaining 12 patients, history of seizure given by guardians was relied upon. Forty-one patients (82%) presented with generalized seizures. Generalized tonic posturing of whole body and up rolling of eye balls was the consistent seizure pattern in all these patients. Only two patients (4%) had focal seizures. Both these patients had unilateral clonic movements of limbs. Seizures in seven patients (14%) could not be classified. Four patients (8%) had acute symptomatic seizure.⁶ Only two patients (4%) were received in status epilepticus. Shock was present in these two patients on admission and one patient developed shock during hospital stay (total 6%). Respiratory failure developed in three patients (6%), needing mechanical ventilation. Family history was significant in 12 patients (24%) with 8 patients (16%) having a first-degree relative with a seizure disorder or having experienced

Table 1 Baseline characteristics

Characteristic	N (%)
Male	31 (62)
Female	19 (38)
Mean age at seizure occurrence (mo)	5.3 ± 2.1
Mean weight (kg)	6 ± 2.5
Mean length (cm)	65.4 ± 10
Mean OFC	46.6 ± 2
Consanguinity	2 (4)
Family history of seizures	12 (24)
Generalized tonic seizures	41 (82)
Focal seizures	2 (4)
Unclassified seizures	7 (14)
Status epilepticus	2 (4)
Shock	3 (6)
Fever	7 (14)
Respiratory failure	3 (6)
Developmental delay at admission (DASII)	4 (8%)
Mean gestational age (wk)	38 ± 1.2
Gross external congenital anomaly	0
Neurocutaneous marker	0

Abbreviations: DASII, Development Assessment Scale for Indian Infants; OFC, occipitofrontal circumference.

seizure episode in the past. Four patients (8%) had a significant history in second-degree relatives. Third-degree consanguinity was noticed in two patients (4%). On examination, none of the patients was found to have a neurocutaneous marker, gross visible external anomaly, or dysmorphic features suggestive of syndromic or chromosomal disorder. Also, none of the patients had any significant organomegaly, visual, or hearing impairment. Four (8%) patients had developmental delay on admission. Baseline characteristics are given in ► **Table 1**. Except for one patient with hypocalcemia, no other patient in the study group was found to have any electrolyte imbalance or hypoglycemia on initial evaluation. Examination of the cerebrospinal fluid was done in all six typical febrile seizure patients and none had meningitis. One patient who had three episodes of seizures along with fever turned out to have meningitis. The remaining 43 patients did not have fever at presentation. Two patients died soon after presenting with seizures before any imaging or EEG could be done. One patient died at 6 months after presenting with first time seizures. EEG and brain imaging were done in 42 patients as indicated (excluding 6 typical febrile seizures and 2 patients who died during initial admission). Among these 42 patients, EEG abnormalities were noted in 15 patients (35.7%), while as imaging abnormalities were noted in 10 patients (23%) (► **Table 2**).

Fifteen patients (35.7%) were found to have developmental delay on follow-up. All 10 patients with abnormal imaging were found to have developmental delay at 6 months follow-

Table 2 Imaging findings

Imaging	No. of patients
Holoprosencephaly	2
Colpocephaly	1
Hydrocephalus	1
Schizencephaly	1
Corpus callosum agenesis	1
CNS infarct	2
Leigh's disease	1
Basal ganglia hyperintensity (infantile beriberi)	1

Abbreviation: CNS, central nervous system.

up. Five patients with developmental delay had normal brain imaging. Among 32 patients with normal development who underwent brain imaging, none had abnormal imaging. The association of developmental delay with abnormal imaging was found statistically significant (p -value < 0.0005). The RR of developing developmental delay at 6 months after first seizure in patients with abnormal imaging was 6.4 times (95% confidence interval [CI] of 2.8–14.3) as compared with those with normal imaging. Only 3 out of 15 patients with abnormal EEG went on to have developmental delay, and the remaining 7 patients with developmental delay had normal EEG (RR: 0.45 with 95% CI of 0.15–1.34). Hence, the risk of developmental delay at 6 months after first seizure in patients with abnormal EEG did not differ significantly from those having normal EEG. Sixteen patients (32%) had seizure recurrence on follow-up. Out of these, nine patients had abnormal imaging, while seven had normal imaging. Out of the remaining 26 patients who did not have recurrence of seizure, only one had abnormal imaging, while 25 had normal imaging giving a RR of 4.1 with 95% CI of 2.1 to 8.1. Among the 16 patients who had seizure recurrence, only 3 patients had an abnormal EEG, and 13 had normal EEG. Out of the remaining 26 patients who did not have recurrence of seizure, 12 had abnormal EEG, and 14 had normal EEG giving a RR of 0.41 with a 95% CI of 0.41 to 1.22. Statistical analysis thus revealed a significant risk of recurrence within first 6 months of first seizure in the presence of abnormal imaging as compared with those with normal imaging; however, the risk of recurrence in the presence of abnormal EEG was not significantly different from those with normal EEG.

There were three deaths (6%) in the study group. Two deaths occurred on initial admission because of shock and multiorgan failure and one death occurred during follow-up at home because of unknown reasons.

Discussion

Our study found a high percentage (42.9%) of children presenting as first seizure between the age of 1 month and 1 year. This is in conformity with other studies which have documented seizure incidence being highest in infancy among pediatric patients.^{2,7} However, these studies have

Table 3 Etiology

Etiology	No. of patients (%)
Simple febrile seizure	6 (12)
Meningitis	1 (2)
Infantile spasms	2 (4)
Hypocalcemia	1 (2)
Structural brain lesions	6 (12)
CNS infarcts	2 (4)
Unprovoked	32 (64)

Abbreviation: CNS, central nervous system.

included neonatal seizures as well. Male patients were found to constitute 62% of total study group, indicating a higher incidence of seizures in males as compared with females in this particular age group. Other studies on pediatric seizures have documented similar result⁸; 82% patients presented with generalized seizures which is a higher percentage when compared with studies done on childhood epilepsies as one group. One study reported 66% patients presenting with generalized seizures in a cohort of 50 up to 3 years of age.⁹ Generalized tonic-clonic seizures were reported as the most common seizure type (69.9%) in children in a study from Nepal.¹⁰ However, our study found generalized tonic seizures as the most common seizure pattern in this particular age group. Although we tried to elicit history related to seizure semiology in detail in cases where seizures were not observed in hospital, there may be a possibility of miscommunication in a few cases.

Our study found a different etiological pattern as compared with few other studies done in this age group. Instead of acute symptomatic seizures (related to central nervous system infections, hypocalcemia, hypoglycemia, electrolyte imbalance, vaccine induced), unprovoked seizures were found to be most common (58%) (► **Table 3**). Also, structural central nervous system abnormalities were noted in a significant number of patients (12%). A recent study reported 51% patients in the same age group presenting as provoked seizures with hypocalcemia constituting 26% of seizures.¹¹ However, like our study, the sample size studied was small. A larger cohort of patients may be needed to arrive at a proper conclusion in this regard. Our study noted EEG abnormalities in 35.7% of patients whose EEG was done. Abnormal EEG was reported in 24.1% in children with generalized seizure in a previous study.¹² Imaging abnormalities were found in 23% of patients imaged in our study. This result conforms to other studies done in children although across the whole pediatric age group.¹³

Our study found developmental delay in 35.7% of patients and seizure recurrence in 32% of patients during a 6-month follow-up. Both developmental delay and seizure recurrence were significantly associated with abnormal brain imaging, while association with abnormal EEG was not statistically significant. These associations have not been studied previously in this age group.

Limitations

A relatively small sample size, short follow-up period for studying seizure recurrence and developmental delay and reliance on caretaker history in many cases about seizure event instead of actual observation or video imaging were major limitations of this study. EEG could not be done sooner than 2 weeks in all patients because of logistic issues which might have underestimated the true number of abnormal EEGs.

Conclusion

Our study found a different etiological pattern of postneonatal infant seizures that differed significantly from previous studies. The difference could be explained by a different sample population studied as compared with previous studies which studied childhood seizures as whole. Also, we recommend a larger cohort of patients to study etiology in this age group. Further, this study found brain imaging as an important tool in predicting seizure recurrence and developmental delay in postneonatal infants presenting with first seizure. Studies with large sample size are required to confirm the findings.

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